

Brain Tumor Detection & Classification

from MRI Images using Classical Machine Learning

*A Multi-Descriptor Feature-Fusion Pipeline with Separability-Driven
Hyper-parameter Selection and Statistical Validation*

Project Report

GLCM · LBP · DWT · HOG · First-order Statistics
SVM · KNN · RF · XGBoost · LightGBM · Logistic Regression · MLP · ExtraTrees

Author: _____

Supervisor / Course: _____

Date: _____

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Abstract

This project builds a complete, reproducible pipeline that classifies brain MRI slices into **three tumor categories** — **Glioma, Meningioma, and Pituitary tumor** — **using classical (non-deep) machine learning**. Rather than learning features end-to-end with a CNN, the pipeline extracts four complementary families of handcrafted descriptors (GLCM texture statistics, Local Binary Patterns, Discrete Wavelet Transform sub-band statistics, and Histogram of Oriented Gradients), augments them with first-order intensity statistics, and fuses everything into a single feature vector.

The work is organised in two phases. **Phase 1** selects the optimal hyper-parameters of every descriptor using three fast, classifier-free class-separability metrics (Fisher Discriminant Ratio, Mutual Information, Davies–Bouldin Index) combined into one composite score, avoiding the cost of training a model for every parameter combination. **Phase 2** extracts features with those optimal settings, cleans and standardises them, and benchmarks eight classifiers (SVM, KNN, Random Forest, XGBoost, LightGBM, Logistic Regression, MLP, ExtraTrees) with exhaustive **GridSearchCV** tuning. A dedicated statistical-validation stage (Cohen’s κ , bootstrap confidence intervals, McNemar, Wilcoxon, Friedman + Nemenyi) then establishes whether the performance differences between models are statistically meaningful.

Keywords:

brain tumor, MRI, texture analysis, GLCM, LBP, DWT, HOG, feature fusion, GridSearchCV, statistical validation, class separability.

1. Problem Statement — Image Classification with Machine Learning

1.1 Clinical motivation

Brain tumors are among the most life-threatening conditions, and the **type of tumor dictates the treatment**. Gliomas, meningiomas, and pituitary tumors differ in origin, location, growth pattern, and prognosis, so distinguishing them from an MRI scan is a decisive early step. In practice this reading is done manually by radiologists, which is **time-consuming, requires scarce expertise, and is subject to inter-observer variability**. An automated classifier can act as a fast, consistent second opinion and a triage aid.

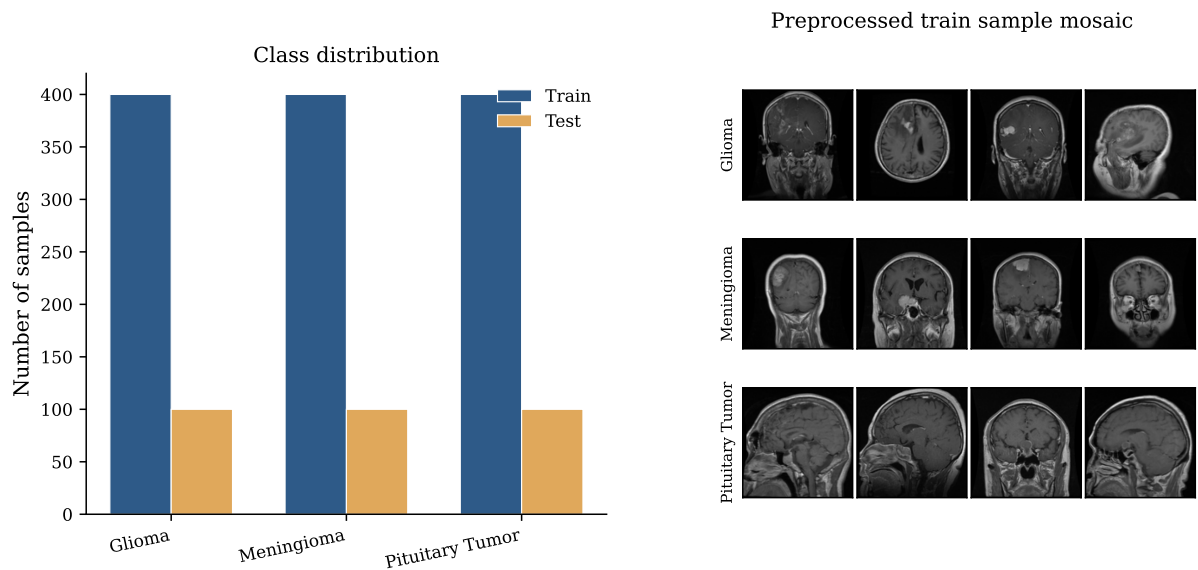


Figure 1: Dataset overview for the classification task. **Left:** class distribution after an 80/20 stratified split (train in blue, test in orange) for the three tumor types. **Right:** mosaic of preprocessed training slices showing acquisition variability in orientation, contrast, and tumor appearance — the visual diversity the classifier must handle.

1.2 Why this is a hard machine-learning problem

Casting tumor identification as an image-classification task exposes several difficulties that the rest of the report is designed to address:

- **High dimensionality.** A modest 128×128 grayscale image already has 16,384 raw pixels. Feeding raw pixels to a classical model invites the curse of dimensionality and overfitting.
- **Subtle, texture-based differences.** Tumor classes are not separated by simple colour or shape cues; the discriminative information lives in fine texture, intensity heterogeneity, and edge structure — properties a single feature type rarely captures alone.
- **Acquisition variability.** MRI slices vary in contrast, orientation, field of view, and the amount of black background around the brain. Without normalisation, the model learns the scanner rather than the disease.
- **Limited, imbalanced data.** Medical datasets are small relative to natural-image datasets; class balance depends on the source corpus (Figure 3). Stratification and macro-averaged metrics are therefore required.

- **The need for trust.** A clinical model must be not only accurate but interpretable and statistically credible — a single accuracy number on one test split is not enough evidence.

1.3 Our design choice: classical ML with handcrafted features

Deep CNNs are the obvious modern tool, but they need large datasets, heavy compute, and behave as black boxes. This project deliberately takes the **classical route**: engineer interpretable descriptors that encode domain knowledge (texture, frequency content, gradient structure), then let well-understood classifiers learn from a compact, meaningful feature space. This is fast to train, runs on a CPU, is fully reproducible, and — crucially for a clinical context — every feature has a clear physical meaning.

Classical ML pipeline for brain-tumor MRI classification

GLCM / LBP / DWT / HOG / Stats

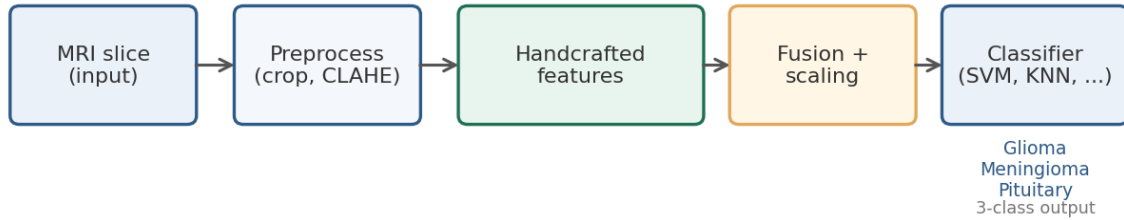


Figure 2: Classical machine-learning pipeline adopted in this project: one preprocessed MRI slice is converted into compact handcrafted descriptors, fused and scaled, then classified by a tuned classical model. This design trades end-to-end deep learning for interpretability, low compute cost, and reproducibility.

Task summary:

Input = one preprocessed MRI slice. Output = one of {Glioma, Meningioma, Pituitary tumor}. Method = handcrafted multi-descriptor features → fusion → tuned classical classifier, validated statistically.

2. Dataset and Preprocessing

2.1 Dataset

The dataset is organised as one folder per class (`images_by_class/<ClassName>/`). After loading and an 80/20 stratified split (random seed 42 for full reproducibility), the working set is:

Property	Value
Classes (3)	Glioma · Meningioma · Pituitary Tumor
Training images	1,200
Test images	300
Total	1,500
Image size after preprocessing	128 × 128, single-channel (grayscale)
Split	80 % train / 20 % test, stratified by class, seed = 42

The split is **stratified** so that each class keeps the same proportion in train and test, and the seed is fixed so the experiment can be repeated exactly. Figure 3 summarises class counts and sample appearance; counts are exported to `figures/data/class_distribution.csv` by `figures/scripts/gen_dataset_figure.py`.

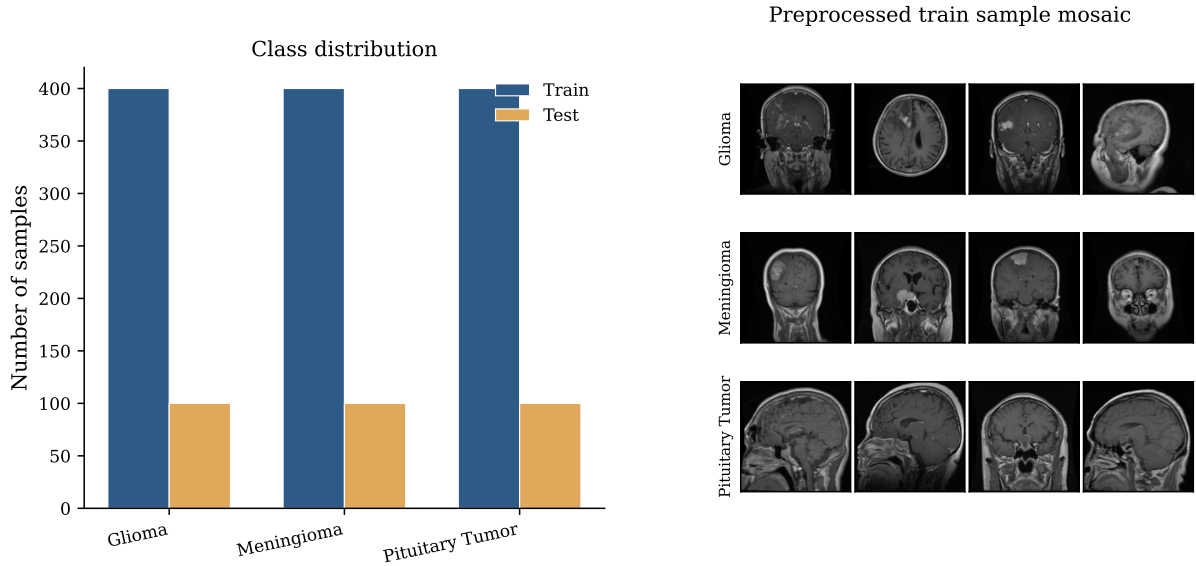


Figure 3: Dataset overview (auto-generated). **Left:** stratified train/test counts per class. **Right:** mosaic of training slices (raw resize preview; full preprocessing in Figure 4). Regenerate with `.\build_figures.ps1`.

2.2 Preprocessing pipeline

Every image passes through an identical normalisation chain before any feature is computed. This removes scanner- and acquisition-related nuisance variation so the descriptors describe anatomy, not artefacts (Figure 4):

1. **Grayscale conversion** — colour channels (if any) are collapsed; MRI intensity is what matters.
2. **Black-border cropping** — the surrounding black region is detected (intensity > 0.05) and cropped away so the brain fills the frame consistently.
3. **Resize to 128×128 with anti-aliasing** — gives every image a common, fixed resolution so feature vectors are comparable.
4. **Gaussian smoothing** ($\sigma = 0.5$) — suppresses high-frequency acquisition noise without blurring real structure.
5. **Min-max normalisation to $[0, 1]$** — puts every image on the same intensity scale.
6. **CLAHE** (Contrast-Limited Adaptive Histogram Equalisation, clip = 0.02) — locally boosts contrast so subtle tumor texture becomes visible to the descriptors.

Why it matters:

Cropping + resizing make features spatially comparable; smoothing + CLAHE make texture descriptors (GLCM, LBP) respond to real tissue patterns instead of noise and global brightness. The whole chain is cached to disk so re-runs are fast.

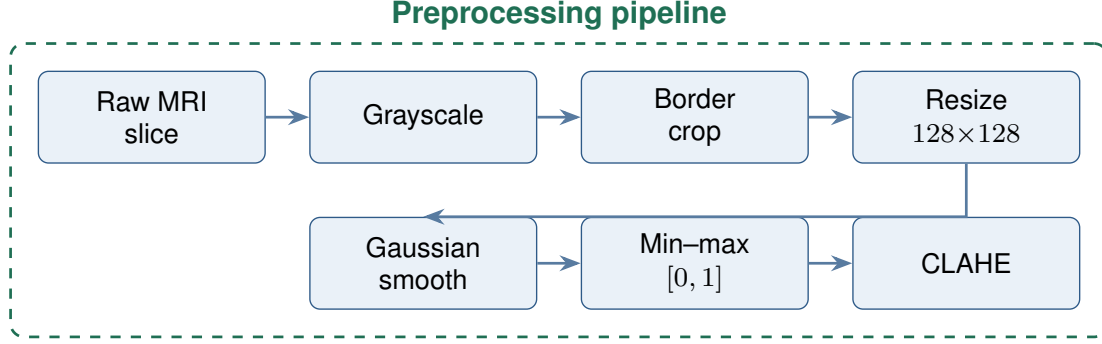


Figure 4: Preprocessing pipeline (TikZ schematic, `figures/tikz/preprocessing_pipeline.tex`). Each slice follows the same deterministic chain before descriptor extraction.

3. Feature-Extraction Strategy

No single descriptor captures everything that distinguishes the three tumor types. The pipeline therefore uses a **multi-branch (multi-descriptor) design**: four complementary feature families are computed independently and then fused. The intuition is that their errors are partly uncorrelated, so the union is more discriminative than any one alone.

Branch	Descriptor	Captures	Question it answers
A — Stats	First-order statistics	Global intensity distribution	How bright / heterogeneous is the tissue overall?
B — Texture	GLCM + LBP	Spatial texture & micro-patterns	How are intensities arranged relative to each other?
C — DWT	Wavelet sub-band stats	Multi-scale frequency content	What structure appears at coarse vs. fine scales?
D — HOG	Gradient orientation hist.	Edge & shape structure	What are the dominant edge directions and contours?

Each branch is extracted with its **own optimal hyper-parameters** (Section 5), concatenated with first-order statistics, and the resulting long vector is cleaned and standardised before modelling. Because the branches are produced separately, the pipeline can also benchmark them individually and in combinations (A, B, C, D, A+B, B+C, A+B+C, and the full fusion) to show exactly how much each contributes.

Design principle:

Diversity over redundancy. GLCM/LBP describe texture, DWT describes scale/frequency, HOG describes shape/edges, and Stats describe global intensity — four different physical views of the same image.

4. The Feature Descriptors in Detail

This section explains what each descriptor measures, the intuition behind it, the parameters that control it, and the size of the vector it produces in this project.

4.1 GLCM — Gray-Level Co-occurrence Matrix

Idea. GLCM is the classic Haralick texture descriptor. It counts how often a pixel of gray level i appears at a fixed offset (distance + direction) from a pixel of gray level j . The result is a co-occurrence matrix $P(i, j)$ that summarises the spatial relationship between intensities — i.e. the **texture** of the tissue. Tumor types differ in how smooth, coarse, or directional their texture is, which GLCM captures well.

What we compute. From each normalised co-occurrence matrix we extract six standard Haralick properties and average them over all distance/angle pairs:

- **Contrast** — local intensity variation (high for coarse texture).
- **Dissimilarity** — like contrast but grows linearly with intensity differences.
- **Homogeneity** — how close $P(i, j)$ mass sits to the diagonal (high for smooth regions).
- **Energy / ASM** — textural uniformity (Angular Second Moment and its square root).
- **Correlation** — linear dependency of gray levels at the given offset.

Parameters tuned.

Parameter	Meaning	Effect
distances	Pixel offsets at which co-occurrence is measured	Small = fine texture; large = coarse structure
angles	Directions (radians) of the offset	Captures directional vs. isotropic texture
levels	Number of gray-level bins	Higher = finer detail but noisier / sparser matrix
symmetric	Average $P(i, j)$ with $P(j, i)$	Makes the descriptor direction-agnostic

Output size: 6 features (always), regardless of parameters, because only the six averaged properties are kept.

4.2 LBP — Local Binary Patterns

Idea. LBP describes **micro-texture**. For every pixel it looks at P neighbours on a circle of radius R , thresholds each neighbour against the centre pixel (1 if brighter, 0 otherwise), and reads the bits as a binary code. The histogram of these codes over the whole image is the feature vector. LBP is powerful because it is **invariant to monotonic gray-scale changes** (lighting/contrast) and is extremely cheap to compute.

Parameters tuned.

Parameter	Meaning	Effect
P	Number of sampling points on the circle	More points = finer texture, longer histogram
R	Radius of the neighbourhood (pixels)	Larger = captures coarser texture
method	LBP variant: uniform / nri_uniform / ror	Controls rotation-invariance & histogram length

The **uniform** and **nri_uniform** variants keep only patterns with ≤ 2 bit-transitions, giving a compact $P+2$ -bin histogram that is robust to noise; **ror** is rotation-invariant but denser. Histograms are normalised to a probability distribution so image size does not affect the values.

Output size: $P + 2$ bins for the uniform methods (so the selected configuration produces **10 features**).

4.3 DWT — Discrete Wavelet Transform

Idea. The DWT decomposes an image into **frequency sub-bands at multiple scales**. A 2-D wavelet transform splits the image into an approximation band (LL — coarse content) and three detail bands (LH, HL, HH — horizontal, vertical, and diagonal edges). Applying it repeatedly to the LL band gives a multi-resolution pyramid. This lets the descriptor see structure that only emerges at a particular scale, which is valuable because tumor tissue and healthy tissue differ in their multi-scale frequency signature.

What we compute. Rather than keep the (large) coefficient images, we summarise every sub-band with four statistics: **mean, standard deviation, energy (mean of squares), and entropy**. A level- L decomposition has $3L + 1$ sub-bands, giving $4 \times (3L + 1)$ features.

Parameters tuned.

Parameter	Meaning	Effect
wavelet	Wavelet family (haar, db2, db4, sym2, coif1, bior1.3 ...)	Sets smoothness & symmetry of the basis
level	Decomposition depth	More levels = coarser bands, more features

Output size: $4 \times (3 \cdot \text{level} + 1)$. The selected configuration (level 1) produces **16 features**.

4.4 HOG — Histogram of Oriented Gradients

Idea. HOG describes **local shape and edge structure**. The image is divided into small cells; within each cell a histogram of gradient orientations (weighted by gradient magnitude) is built; cells are grouped into blocks and normalised for contrast invariance. The concatenation of all block histograms is the feature vector. HOG is excellent at encoding the contours and directional edges that outline a tumor mass.

Parameters tuned.

Parameter	Meaning	Effect
orientations	Number of gradient-direction bins	More bins = finer angular resolution
pixels_per_cell	Spatial size of each cell	Smaller = more detail, much longer vector
cells_per_block	Block size for normalisation	Controls contrast-normalisation neighbourhood
block_norm	Normalisation scheme (L2-Hys)	Robustness to local contrast changes

Output size: depends on cell/block geometry; the selected configuration produces **1,176 features** — by far the largest branch, which is why feature cleaning later matters most for HOG.

Figure placeholder:

Insert the per-descriptor visualisations the notebook saves (e.g. the HOG visualisation image and the GLCM property bar chart) here to make this section concrete.

5. Selecting the Best Hyper-parameters per Descriptor (Phase 1)

5.1 The problem with brute force

Each descriptor has many possible settings. Training a classifier for every combination would take hours and conflate **feature quality** with **model quality**. We want to judge the features themselves, quickly and fairly.

5.2 Classifier-free class-separability metrics

Instead, every candidate configuration is scored in milliseconds with three complementary metrics that measure how well the resulting feature distributions separate the three classes. All features are standardised first so the comparison is fair across configurations.

Metric	What it measures	Direction
Fisher Discriminant Ratio (FDR)	Between-class variance $\div$ within-class variance, averaged over features. Large means class means are far apart relative to their spread.	Higher is better
Mutual Information (MI)	Statistical dependency between each feature and the class label (averaged). Captures non-linear relationships FDR misses; scale-invariant.	Higher is better
Davies–Bouldin Index (DBI)	Ratio of within-class scatter to between-class distance in (PCA-reduced) feature space. Measures cluster compactness & separation.	Lower is better

5.3 Combining them into one score

The three metrics are on different scales, so for each descriptor’s set of candidates they are **min–max normalised to** $[0, 1]$, the DBI is inverted (so that higher is always better), and the three normalised values are averaged:

$$\text{score} = \frac{\text{FDR}_{\text{norm}} + \text{MI}_{\text{norm}} + (1 - \text{DBI}_{\text{norm}})}{3}.$$

The configuration with the highest composite score is selected for each descriptor. This composite is robust: a configuration must do well on **all three** complementary criteria to win, not just one.

5.4 Search space explored

Descriptor	Configurations evaluated
GLCM	6 distance sets $\times$ 4 angle sets $\times$ 3 levels $\times$ 2 symmetry = 144
LBP	4 $P \times 3 R \times 3$ methods = 36
DWT	7 wavelets $\times$ 3 levels = 21
HOG	3 orientations $\times$ 2 cell sizes $\times$ 2 block sizes $\times$ 1 norm = 12
Total	213 configurations, all scored without training a classifier

5.5 Selected optimal parameters (results)

Running Phase 1 on the training set produced the following winners. These are the actual values stored in the notebook:

Descriptor	Selected parameters	Score	Size
GLCM	distances = [1, 2, 4, 8]; angles = [0]; levels = 128; symmetric = True	0.7809	6
LBP	$P = 8$; $R = 1$; method = uniform	1.0000	10
DWT	wavelet = bior1.3; level = 1	0.9941	16
HOG	orientations = 6; pixels_per_cell = (16, 16); cells_per_block = (2, 2); block_norm = L2-Hys	0.9969	1176

Reading these results: **LBP** achieved a perfect composite score (1.0000) on the training set, indicating its uniform $P=8/R=1$ histograms separate the three classes extremely cleanly; **DWT (bior1.3)** and **HOG** are also very strong. **GLCM** scores lower (0.7809) — its six averaged properties carry useful but less individually discriminative information — yet it is still retained because it is cheap and adds a complementary texture view to the fusion. The chosen GLCM offsets [1, 2, 4, 8] deliberately span fine to coarse texture, while a single angle (0) keeps the descriptor compact.

Figure placeholder:

Insert `glcm_param_search.png`, the LBP plots, `dwt_param_search.png`, `hog_param_search.png`, and `phase1_summary.png` from the `output/` folder to visualise the search and the per-descriptor comparison.

6. Building Optimal Features and Benchmarking Models (Phase 2)

With the optimal parameters fixed, Phase 2 carries out the five steps quoted in the project brief: (1) extract features with the best GLCM/LBP/DWT/HOG settings, (2) add first-order statistics, (3) fuse all branches, (4) run `GridSearchCV` per model across eight classifiers, and (5) produce a final performance dashboard.

6.1 First-order statistics (Branch A)

Before fusion, an extra branch of 11 global intensity statistics is computed per image — these have no tunable parameters and give the model a cheap summary of overall tissue appearance: `mean`, `std`, `variance`, `entropy`, `skewness`, `kurtosis`, and the 10/25/50/75/90 intensity percentiles.

6.2 Fusing the branches

All branches are concatenated into one feature vector per image. The dimensions are:

Branch	Descriptor(s)	Features
A	First-order statistics	11
B	Texture = GLCM (6) + LBP (10)	16
C	DWT (bior1.3, level 1)	16
D	HOG	1,176
FULL	Fused A + B + C + D	1,219

To quantify each branch’s contribution, eight feature sets are benchmarked: **A**, **B**, **C**, **D**, **A+B**, **B+C**, **A+B+C**, and **Full(opt)**.

6.3 Cleaning and scaling

Each feature set passes through an identical cleaning pipeline so that the classifier sees only informative, non-redundant inputs:

1. **Variance threshold** — constant (zero-variance) features are removed.
2. **Correlation filter** — for sets under 5,000 dimensions, one of any pair of features with $|\text{correlation}| > 0.95$ is dropped to reduce redundancy.
3. **Standardisation** — every remaining feature is z -scored using statistics fit on the training set only, then applied to the test set, preventing data leakage.

6.4 Model benchmark with GridSearchCV

Eight classifiers are tuned independently. For each, an exhaustive **GridSearchCV** searches its hyper-parameter grid using **5-fold StratifiedKFold cross-validation** and optimises **macro-F1** (chosen over accuracy because it weights all three classes equally despite imbalance). The best estimator is refit on the full training set and evaluated on the held-out test set.

Model	Key hyper-parameters searched
SVM	kernel (linear/RBF/poly/sigmoid), C , γ , degree, coef0
KNN	n_neighbors, distance metric, weights, p
Random Forest	n_estimators, max_depth, max_features, split/leaf sizes, bootstrap, criterion
XGBoost	n_estimators, max_depth, learning_rate, subsample, colsample_bytree, min_child_weight, gamma
LightGBM	n_estimators, max_depth, learning_rate, num_leaves, subsample, colsample_bytree, min_child_samples
Logistic Regression	C , penalty (l1/l2/elasticnet), class_weight, l1_ratio (scaler + saga solver)
MLP	hidden_layer_sizes, activation, alpha, learning_rate_init, batch_size (early stopping)
ExtraTrees	n_estimators, max_features, split/leaf sizes, criterion

Beyond the held-out split, the pipeline also records cross-validation evaluation modes (**cv_1...cv_5**) so that per-fold scores are available for the statistical tests in Section 7. Every (model $\times$ feature-set) job is cached to disk, making the long benchmark resumable.

6.5 Benchmark results

Fill-in — edit `results.tex`:

The numbers below are pulled from macros in `results.tex`. Replace the placeholders there with the values printed by the `GridSearchCV/finalize` cells, then recompile.

Model	Best CV F1	Test Acc	Test F1	Best params (summary)
SVM	[CV]	[Acc]	[F1]	[params]
KNN	[CV]	[Acc]	[F1]	[params]
Random Forest	[CV]	[Acc]	[F1]	[params]
XGBoost	[CV]	[Acc]	[F1]	[params]
LightGBM	[CV]	[Acc]	[F1]	[params]
Logistic Reg.	[CV]	[Acc]	[F1]	[params]
MLP	[CV]	[Acc]	[F1]	[params]
ExtraTrees	[CV]	[Acc]	[F1]	[params]

Headline result. The best configuration was [best model] on the [best feature set] feature set, reaching a test accuracy of [acc] and a macro-F1 of [macro-F1].

Figure placeholder:

Insert the performance dashboard (grouped F1 bar chart, F1 heatmap, training-time vs. F1 scatter) and the leaderboard chart produced by Sections 13–14 of the notebook.

7. Statistical Validation

A single accuracy number on one test split is weak evidence: it could be luck. This stage establishes whether the observed differences between models are **real, stable, and reproducible**. Six complementary tests are applied, moving from per-model agreement to rigorous multi-classifier comparison.

7.1 Cohen’s Kappa (κ)

Measures agreement between predictions and ground truth **beyond chance**, correcting for class imbalance (so it is stricter than accuracy). Interpretation: < 0.20 slight, 0.20 – 0.40 fair, 0.40 – 0.60 moderate, 0.60 – 0.80 substantial, > 0.80 almost perfect. Computed for every (feature-set $\times$ model) on the test split.

7.2 Bootstrap Confidence Intervals (95 %)

The test predictions are resampled with replacement $N = 2,000$ times; the 2.5th and 97.5th percentiles of the resulting metric distribution form a **95 % confidence interval** around accuracy

and macro-F1. Non-overlapping intervals between two models are strong evidence of a genuine difference.

7.3 Cross-validation fold statistics

The mean $\pm$ standard deviation of macro-F1 across the CV folds quantifies **stability**: a model with a high mean but large variance is less trustworthy than a slightly lower but consistent one.

7.4 McNemar’s test (pairwise, same test set)

The standard significance test for two classifiers evaluated on the **same** samples. It builds a contingency table of where the two models disagree (A right/B wrong vs. A wrong/B right) and tests, with continuity correction, whether those disagreements are balanced. $p < 0.05$ means the two classifiers make significantly different errors. Run across all model pairs on the best feature set, `Full(opt)`.

7.5 Wilcoxon signed-rank test (paired, across folds)

A non-parametric paired test comparing two classifiers’ per-fold macro-F1 scores. It is preferred over a paired t -test because, with only five folds, normality cannot be assumed. H_0 : the median per-fold difference is zero (the models are equivalent).

7.6 Friedman test + Nemenyi post-hoc

The gold standard for comparing **many** classifiers over multiple folds (Demšar, 2006). The Friedman test (a non-parametric repeated-measures ANOVA) first asks whether **any** classifier differs from the others; if it is significant ($p < 0.05$), the **Nemenyi post-hoc** test identifies exactly which pairs differ while controlling for multiple comparisons.

7.7 Statistical summary (results)

Fill-in — edit `results.tex`:

These tests depend on the completed Phase 2 predictions. Replace the placeholders in `results.tex` once the benchmark finishes.

Test	Model / pair	Statistic	p -value	Conclusion
Cohen’s κ (best)	[model]	$\kappa = [\text{kappa}]$	—	[interpretation]
Bootstrap 95 % CI (F1)	[model]	[[lo, hi]]	—	—
CV fold mean $\pm$ std	[model]	[mean] $\pm$ [std]	—	—
McNemar (top pair)	[A vs B]	$\chi^2 = [\text{chi2}]$	[p]	[sig / not sig]
Wilcoxon (top pair)	[A vs B]	$W = [\text{W}]$	[p]	[sig / not sig]
Friedman (<code>Full(opt)</code>)	all models	$\chi^2 = [\text{chi2}]$	[p]	[sig / not sig]

Figure placeholder:

Insert the Kappa heatmap, the bootstrap-CI plot for the best feature set, the CV mean $\pm$ std bars, the Nemenyi heatmap, and the final CI strip-plot.

8. Results and Discussion

Write this section **after** filling `results.tex`. Use the prompts below so the discussion reads as analysis, not just a list of figures.

- **Best overall configuration.** State which (feature-set $\times$ model) pair won on test macro-F1 and by how much, and confirm whether the full fusion beat the individual branches — quantify the gain.
- **Branch contribution.** Compare A vs B vs C vs D and the cumulative sets. Did HOG dominate (it is the largest branch)? Did adding GLCM/LBP texture or DWT measurably help over HOG alone?
- **Accuracy vs. cost.** Reference the training-time-vs-F1 scatter: if a lighter model (e.g. SVM or LightGBM) matches the best within its confidence interval, argue for it on efficiency grounds.
- **Statistical significance.** Tie the leaderboard to Section 7: are the top models statistically separable (McNemar / Friedman+Nemenyi), or is the top cluster a statistical tie? This is the most important point a grader looks for.
- **Per-class behaviour.** From the confusion matrices, note which tumor types are confused most (commonly Glioma $\leftrightarrow$ Meningioma) and relate it back to the texture similarity discussed in Section 1.

8.1 Threats to validity

- Single dataset and a single train/test split (mitigated by stratification, fixed seed, and CV — a seed-stability check across three seeds is included in the notebook diagnostics).
- Phase 1 selects parameters on the training set using separability metrics, not downstream accuracy; the metrics are a fast proxy, so the chosen settings are good but not guaranteed globally optimal for every classifier.
- Handcrafted features cannot adapt the way a CNN would; the trade-off is interpretability, speed, and low data requirements.

9. Conclusion

This project delivered an end-to-end, reproducible, and interpretable pipeline for classifying brain MRI slices into glioma, meningioma, and pituitary tumor using only classical machine learning. Its main contributions are a **multi-descriptor feature design** (GLCM, LBP, DWT, HOG, plus first-order statistics) that captures texture, multi-scale frequency, and edge structure; a **fast, classifier-free hyper-parameter selection** stage that scores 213 descriptor configurations with three complementary separability metrics; an **exhaustive eight-model benchmark** tuned by GridSearchCV; and a **thorough statistical-validation** stage that tests whether the results are genuinely meaningful rather than noise.

Phase 1 cleanly selected optimal settings for every descriptor (LBP scoring a perfect composite separability of 1.0000, with DWT and HOG close behind), and the fused feature vector of 1,219 dimensions provides a rich, complementary representation for the model benchmark.

9.1 Future work

- Validate on an independent external dataset and with patient-level (not slice-level) splitting to rule out leakage.
- Add a learned feature-selection or dimensionality-reduction step (e.g. recursive feature elimination) to shrink the 1,176-dim HOG branch.
- Compare against a lightweight CNN baseline to quantify the interpretability-vs-accuracy trade-off.
- Build the best statistically-tied ensemble (the notebook already includes a soft-voting SVM+XGBoost+Light experiment) for a final deployable model.

9.2 Reproducibility checklist

- Fixed random seed (42) for splitting, models, and bootstrap.
- Stratified 80/20 split; 5-fold StratifiedKFold; macro-F1 scoring throughout.
- Disk caching of the dataset, Phase-1 scans, fused features, and each benchmark job, so the whole study can be regenerated exactly.